Part I

A. Qualitative Observations and Spectroscope Measurements

Table 1: Data for hydrogen lamp emission lines using a spectroscope

Report the visible emission lines in order of **decreasing wavelength**.

Observed line	Wavelength (nm)	Color	Relative intensity
Line 1	655	Red	very bright
Line 2	490.	Blue	very bright
Line 3	440.	Violet (purple)	dim

Discussion Q1

In Table 1, which wavelength represents the electron transition with smallest change in energy?

655 nm

Explain the rationale for your conclusion.

The red line at 655 nm has the longest wavelength of the three lines. Since the energy of a photon is E = hc/λ, a longer wavelength means less energy, so the 655 nm line represents the electron transition with the smallest change in energy (ni = 3 to nf = 2).

Table 3: Hydrogen lamp emission lines from a USB spectrometer

Report the visible emission lines in order of **decreasing wavelength**.

(Rounded $^{*}1/(n_i)^{2*}$ value should have the same number of sig figs as the wavelength data value.)

Measured wavelength (nm)	1/λ (x10^{-3} nm^{-1})	n_i	$1/(n_i)^2$
661.1	1.513 (1.512630)	3	0.1111 (0.111111)
485.1	2.061 (2.061431)	4	0.06250 (0.0625000)
432.6	2.312 (2.311604)	5	0.04000 (0.0400000)
409.0	2.445 (2.444988)	6	0.02778 (0.0277778)

Discussion Q2

Calculate the percent difference between the predicted and measured wavelengths for the next Balmer line.

% difference (Record to 3 sig figs)	0.938

The literature value for the longest visible hydrogen line is 656.3 nm.

(Record "% Error" to 3 sig figs)

Calculation	% Error
Spectroscope longest-wavelength line	0.198
USB spectrometer longest-wavelength line	0.731

Compare the two percent errors and discuss how the instruments affect the quality of the wavelength measurements.

The spectroscope had a lower percent error (0.198%) than the USB spectrometer (0.731%) for the longest line. The spectroscope is read by eye off a printed scale, so the value depends on where we judged the line to be and only has about 3 sig figs. The spectrometer reads intensity at each wavelength and gives a precise peak value, so it is more precise and less subjective, but its peaks were broad and it may be slightly off in calibration, which made its value less accurate for this line.

Hand-drawn plot values

Enter the slope and y-intercept from your hand-drawn plot, then use them to calculate R_H.

Hand-drawn slope ($x10^{-3}$ nm^{-1}) (Record to 4 sig figs)	-11.20
Hand-drawn y-intercept ($x10^{-3}$ nm^{-1}) (Record to 4 sig figs)	2.850
R_H from slope (m^{-1})	1.120e7 (1.1200e7)
R_H from y-intercept (m^{-1})	1.140e7 (1.1400e7)
Average R_H from hand-drawn plot (m^{-1})	1.130e7 (1.1300e7)

Part I Summary

Discussion Q3

Compare your three average R_H values to the literature value of 1.09677581e+7 m^{-1}.

(Record "% Error" to 3 sig figs)

Method	Average R_H (m^{-1})	% Error
Spectroscope single-point average	1.08997e+7	0.621
Hand-drawn plot average	1.1300e+7	3.03
LabRight plot average	1.11187e+7	1.38

Discuss the benefits and drawbacks of the three approaches for determining the value for R_H (i.e., single data point, slope of plot, y-intercept of plot). Which method would you recommend to someone hoping to get the most accurate value? Explain.

A single data point is quick, but it relies on one reading, so any error in that line goes straight into RH. The slope of a plot uses all four lines, so random errors in each reading tend to average out. The y-intercept also uses all the data, but it is extrapolated to 1/ni^2 = 0, so a small error in the line gets magnified, which is why our estimated hand-drawn intercept pushed that average to 3.03% error. Our spectroscope average was closest (0.621%) and the LabRight plot was 1.38%, likely because the spectrometer readings were slightly off. I would still recommend the slope of a computer fit of spectrometer data for the most accurate value, since it uses every line and does not depend on judging by eye, as long as the spectrometer is calibrated.

Table 4: Flame test observations

Sample solution	Color of solution	1st flame color	Is there a 2nd flame color?	2nd flame color
$CuCl_2 \cdot H_2O$	blue-green	green	no	N/A
$BaCl_2$	colorless	green	no	N/A
LiCl	colorless / yellowish	red	no	N/A
$SrCl_2$	colorless / yellowish	red	no	N/A
KCl	colorless / yellowish	purple	no	N/A
KNO_3	yellowish	purple	no	N/A
Unknown A	blue-green	green	no	N/A
Unknown B	blue-green	red	yes	green
Unknown C	colorless / yellowish	purple	no	N/A

Discussion Q4

During lab you compared your flame-test observations with two other groups. Did all three groups report identical flame colors for all samples?

yes

Explain any differences you observed.

We compared with stations 10 and 6. All three groups reported the same flame colors for every sample and identified the same metal ions. The only small difference was how long some flames lasted, since duration is judged by eye and depends on how much solution was on the wire, but it did not change any colors or conclusions.

Discussion Q5

Based on your experience in this lab, suggest a better method for determining the identity of unknown metal-ion solutions.

A better method would be to use a spectrometer to record the emission spectrum of each unknown and compare its line wavelengths to the known metal ions. This is better because it is quantitative and does not depend on judging colors by eye, and it can pick out a second ion even when one flame color covers another, like the green from Cu2+ in Unknown B. The downside is that it takes more time and needs more equipment than a quick flame test.

Laboratory Waste Evaluation

Laboratory waste is considered anything generated during an experiment that is disposed of down the sewer drain, thrown in the garbage, collected in a container for disposal by the UW Environmental Health & Safety department, or released into the environment. Based on the written lab procedure and your actions during the lab, list the identity and approximate amount (mass or volume) of waste that you generated while performing this experiment.

No waste was generated.